Put **2** in front of HCl:

$H_2 + Cl_2 \rightarrow 2HCl$

Balanced equation:

$H_2 + Cl_2 \rightarrow 2HCl$

Example 3: Combustion reaction

Unbalanced equation:

$CH_4 + O_2 \rightarrow CO_2 + H_2O$

Step 1: Count atoms

Left:

- C = 1
- H = 4
- O = 2

Right:

- C = 1
- H = 2
- O = 3

Step 2: Balance hydrogen

Put **2** in front of H_2O:

$CH_4 + O_2 \rightarrow CO_2 + 2H_2O$

Step 3: Balance oxygen

Right:

Put **2** in front of O_2:

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Balanced equation:

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Example 4: Ionic compound formation

Unbalanced equation:

$Al + O_2 \rightarrow Al_2O_3$

Step 1: Count atoms

- **Left:**

  Al = 1

  O = 2
- **Right:**

  Al = 2

  O = 3

Step 2: Balance aluminium and oxygen

Al_2O_3 has **2** Al, so put **2** in front of Al

$2Al + O_2 \rightarrow Al_2O_3$

Al_2O_3 has **3 O**, O_2 has **2 O**

LCM of 2 and 3 = **6**

Put:

- **3** in front of O_2
- **2** in front of Al_2O_3

$2Al + 3O_2 \rightarrow 2Al_2O_3$

Step 3: Balance aluminium again

4 Al on the right , so put **4** in front of Al on the left

$4Al + 3O_2 \rightarrow 2Al_2O_3$

Balanced equation:

$4Al + 3O_2 \rightarrow 2Al_2O_3$

Example 5: Acid + metal reaction

Unbalanced equation:

$Zn + HCl \rightarrow ZnCl_2 + H_2$

Step 1: Count atoms

Left:

- Zn = 1
- H = 1
- Cl = 1

Right:

- Zn = 1
- H = 2
- Cl = 2

Step 2: Balance H and Cl

Put **2** in front of HCl:

$Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Balanced equation:

$Zn + 2HCl \rightarrow ZnCl_2 + H_2$

1.2. Relative Masses of Atoms and Molecules

- **Relative Atomic Mass (A_r):** the average mass of the isotopes of an element compared to 1/12th of the mass of an atom of 12C.
- **Relative Molecular Mass (M_r):** sum of relative atomic masses of all the atoms in one molecule of the compound.

1.3. The Mole and the Avogadro Constant

A **mole** of a substance is the amount that contains the same number of units as the number of carbon atoms in 12 grams of carbon-12

- A mole is the A_r or M_r expressed in grams e.g. 1 mole of Carbon-12 is equal to 12 grams.
- It is equal to 6.02×10^{23} particles; this number is called **Avogadro's constant**.
- 1 mole also occupies a volume of $24dm^3$ at room temperature and pressure

1.4. Number of Moles

$$\text{Amount of Substance} = \frac{\text{mass}}{\text{molar mass}}$$

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